

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

PRINCIPLES OF CELLULAR & MOBILE COMMUNICATIONS

(Common to EEE, CSE, IT, AI&DS, CSE (AI), CSE(DS), CSE(AI&ML), CSE(IOT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
- | | |
|---|----|
| (a) Why are cellular mobile communication systems necessary? | 2M |
| (b) How mobile radio communication evolved over time? | 2M |
| (c) What is cell splitting? | 2M |
| (d) What is the general description of the problem in cellular communication? | 2M |
| (e) What are the different types of handoffs? | 2M |
| (f) What is a soft handoff? | 2M |
| (g) What is Frequency Division Multiple Access (FDMA)? | 2M |
| (h) What is Time Division Multiple Access (TDMA)? | 2M |
| (i) What is the introduction to the 5G system? | 2M |
| (j) How does GSM handle data and voice communication? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Compare analog and digital cellular systems with at least 10 points each. 10M
- OR**
- 3 What is the uniqueness of mobile radio environment? Explain why hexagonal shaped cells are used for mobile communication. 10M
- 4 (a) For a seven cell reuse pattern find the reuse factor if the minimum distance between centers of co-channel-cells is 18 km. Radius of cell is 3 km and the distance between adjacent cells in the seven cell pattern is 6 km. 5M
- (b) What is cell-splitting? Explain its types in detail. 5M
- OR**
- 5 What are the different types of interference for a cellular system in detail? Also Explain the types of non-co-channel interferences in cellular system. 10M
- 6 (a) What are the various methods of delaying handoff? Explain briefly. 5M
- (b) What is meant by dropped call? Explain the factors that influenced dropped call rate. 5M
- OR**
- 7 Explain MAHO and soft handoff techniques. 10M
- 8 (a) Explain the architecture of TDMA. 5M
- (b) Explain in detail about multiple access schemes. 5M
- OR**
- 9 What is Space Division Multiple Access (SDMA), and how does it utilize spatial resources to manage multiple users in wireless communication? 10M
- 10 Compare the key characteristics and technologies used in 2G, 3G, and 4G systems. 10M
- OR**
- 11 Draw the GSM architecture and discuss various interfaces used in GSM. 10M

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Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|---|----|
| (a) Write about basic cellular system. | 2M |
| (b) What is digital cellular system? | 2M |
| (c) Define frequency reuse factor. | 2M |
| (d) What is cell sectoring? | 2M |
| (e) Define Handoff. | 2M |
| (f) What is intersystem Handoff? | 2M |
| (g) Define efficiency in TDMA. | 2M |
| (h) Write the advantages of CDMA. | 2M |
| (i) What are the standards used in 2G technology? | 2M |
| (j) Write the advantages of 4G system. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | |
|---|----|
| 2 (a) Describe the principle of operation of cellular mobile system. | 5M |
| (b) Explain the evolution of mobile radio communication. | 5M |
| OR | |
| 3 (a) Discuss about characteristics of mobile radio environment. | 5M |
| (b) Explain briefly analog cellular mobile system. | 5M |
| 4 (a) Discuss about frequency reuse schemes. | 4M |
| (b) Explain briefly cell splitting and sectoring. | 6M |
| OR | |
| 5 (a) Discuss about co-channel interference reduction factor. | 5M |
| (b) Explain about derived C/I ratio. | 5M |
| 6 (a) Explain about Handoff process in cellular mobile communication. | 5M |
| (b) Explain about cell-site handoff. | 5M |
| OR | |
| 7 (a) Explain dropped calls and dropped call rate. | 5M |
| (b) Compare between mobile assisted handoff and soft handoff. | 5M |
| 8 (a) Draw the structure and explain the operation of FDMA. | 6M |
| (b) Comparison between FDMA and CDMA. | 4M |
| OR | |
| 9 (a) Draw the structure and explain the operation of TDMA. | 4M |
| (b) Write advantages, disadvantages and applications of SDMA. | 6M |
| 10 (a) Explain in detail Global system for mobile systems. | 5M |
| (b) Write a brief note on 5G system. | 5M |
| OR | |
| 11 (a) Explain TDMA-based 2G standards. | 5M |
| (b) Discuss about multi accesses in CDMA. | 5M |
